

Geospatial Visualization with R

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Why Geospatial Visualization Matters

- Understand spatial patterns
- Support decision-making
- Explore relationships between places
- Transform data into visual insight

What is Geospatial Data?

Geospatial data is information linked to a location.

Examples:

- Roads
- Buildings
- Cities
- Countries
- Rivers

Vector vs Raster Data

Vector	Raster
Points	Cells
Lines	Pixels
Polygons	Continuous surfaces
Roads	Elevation
Buildings	Temperature

Layer Concept

A map is built from multiple layers.

Examples:

- Country boundaries
- Cities
- Roads
- Population

Layers can be combined to answer spatial questions.

Why Use R?

R provides:

- Reproducible workflows
- Data transformation
- Visualization
- Spatial analysis
- Open-source tools

Packages

```
1 library(sf)
2 library(tidyverse)
3 library(ggplot2)
4 library(rnaturalearth)
5 library(rnaturalearthdata)
```

Load Spatial Data

```
1 world <- ne_countries(  
2   scale = "medium",  
3   returnclass = "sf"  
4 )  
5  
6 germany <- world |>  
7   filter(admin == "Germany")
```

Simple feature collection with 1
feature and 168 fields

Geometry type: MULTIPOLYGON

Dimension: XY

Bounding box: xmin: 5.85752 ymin:
47.27881 xmax: 15.0166 ymax: 55.05874

Geodetic CRS: WGS 84

	featurecla	scalerank	labelrank
sovereignty	sov_a3	adm0_dif	level
1	Admin-0	country	1 2
Germany	DEU	0	2
	type	tlc	admin
adm0_a3	geou_dif	geounit	gu_a3 su_dif
subunit			

Coordinate Reference System

```
1 st_crs(germany)
```

Coordinate Reference System:

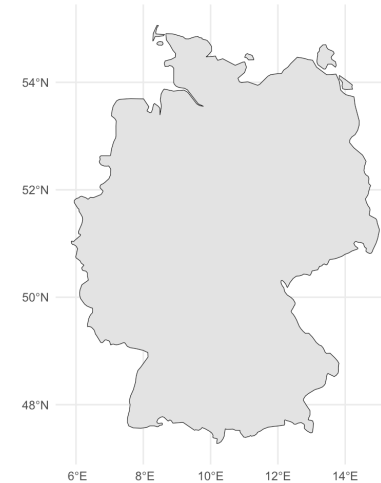
User input: WGS 84

wkt:

```
GEOGCRS["WGS 84",  
    DATUM["World Geodetic System  
1984",  
        ELLIPSOID["WGS  
84",6378137,298.257223563,  
            LENGTHUNIT["metre",1]],  
    PRIMEM["Greenwich",0,  
        ANGLEUNIT["degree",0.0174532925199433]]  
    CS[ellipsoidal,2],  
    AXIS["latitude",north,  
        -- -- --
```

Basic Vector Map

```
1 ggplot() +  
2   geom_sf(  
3     data = germany,  
4     fill = "grey90",  
5     color = "black"  
6   ) +  
7   theme_minimal()
```



A Question You Should Have Asked

Before creating a map:

“How do we know what is inside this dataset?”

We should never use a dataset blindly.

Useful functions:

`names(xxxx)`

`glimpse(xxxx)`

These functions help us discover:

- available variables
- data types
- relevant columns
- possible filters

Only then we can decide what to visualize.

In-Class Activity: Mapping German Cities

In this activity, we will build a map step by step.

Goal:

- add city points
- combine spatial layers
- filter cities by population
- interpret the result

Activity Step 1: Add Cities

```
1 cities <- ne_download(  
2   scale = 10,  
3   type = "populated_places",  
4   category = "cultural",  
5   returnclass = "sf"  
6 )  
7  
8 germany_cities <- cities |>  
9   filter(ADM0NAME == "Germany")
```

Simple feature collection with 8 features and 2 fields

Geometry type: POINT

Dimension: XY

Bounding box: xmin: 6.750017 ymin: 48.13189 xmax: 13.3996 ymax: 53.55197

Geodetic CRS: WGS 84

NAME POP_MAX

geometry

1 Berlin 3406000 POINT (13.3996 52.52376)

2 Stuttgart 2944700 POINT (9.199996 48.77998)

3 Frankfurt 2895000 POINT (8.675015

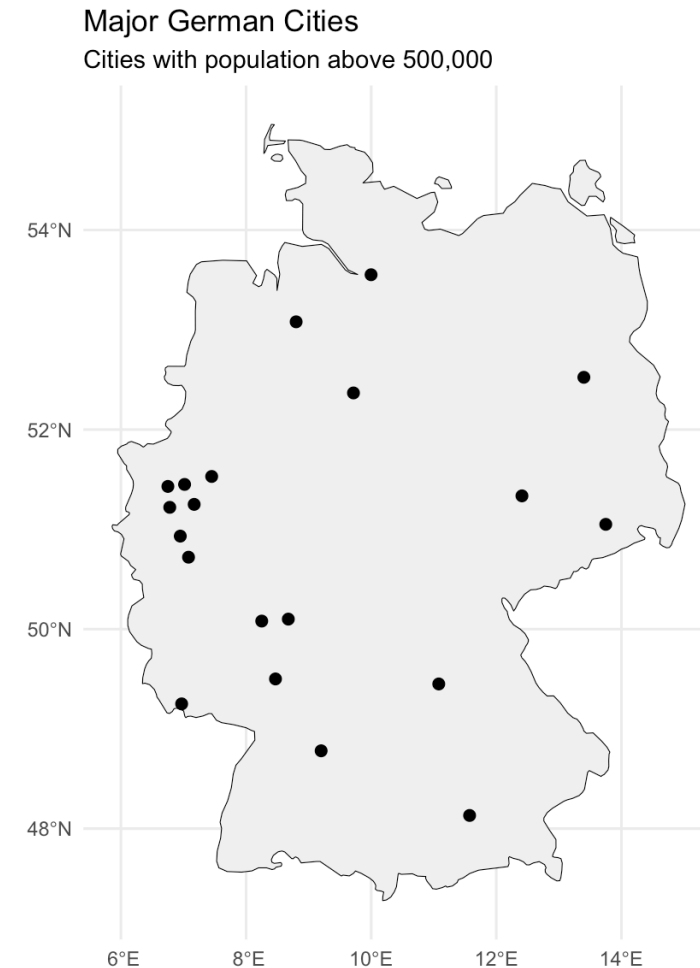
Activity Step 2: Combine Spatial Layers

```
1 ggplot() +  
2   geom_sf(  
3     data = germany,  
4     fill = "grey95",  
5     color = "black"  
6   ) +  
7   geom_sf(  
8     data = germany_cities,  
9     size = 0.8,  
10    alpha = 0.5  
11  ) +  
12  theme_minimal()
```



Activity Step 3: Filter the Dataset

```
1 germany_cities_major <- germany_cit
2   filter(POP_MAX > 500000)
3
4 ggplot() +
5   geom_sf(
6     data = germany,
7     fill = "grey95",
8     color = "black"
9   ) +
10  geom_sf(
11    data = germany_cities_major,
12    size = 2
13  ) +
14  theme_minimal()
```



Activity Step 4: Interpret the Map

Discussion questions:

- What changed after filtering?
- Why is filtering useful before visualization?
- Which cities remain visible?
- How can this support decision-making?

Key Takeaways

- Geospatial data connects information to location.
- Vector and raster data serve different purposes.
- Layers can be combined.
- R provides reproducible geospatial workflows.

References

Lovelace, R., Nowosad, J., & Muenchow, J. (2019). *Geocomputation with R*.

Pebesma, E., & Bivand, R. (2023). *Spatial Data Science with Applications in R*.

Huber, S. (n.d.). *How to Use R for Data Science*.

Questions?

Thank you.